Image_projet_money
Coin Detection and Value Estimation
Master 1 – VMI Analyse d'Image

Maksym Dolhov, Mehdi Aghaei, Nima Davari

Universite Paris Cite

February 24, 2026

GitHub: `github.com/sorooshaghaei/Image_projet_money`

Presentation Plan (15 minutes)

Project Goal

- Input: one RGB image with multiple euro coins on different surfaces and lighting conditions.
- Output 1: number of detected coins.
- Output 2: estimated total value (in EUR/cents).
- Constraint: use classical computer vision + deterministic heuristics from our implemented codebase.
- Main code references: `main.py`, `src/`, `PIPELINE.md`, `README.md`.

Dataset and Evaluation Protocol

- Dataset folder: `data/images/` with groups gp1 ... gp8.
- Ground truth source: `src/data/ground_truth.py` (`RAW_ANNOTATIONS` table).
- Processed images: 106.
- Evaluated with ground truth: 105 (1 image skipped: missing value/annotation row).
- Metrics computed by `src/evaluation/metrics.py`; report rows written by `src/evaluation/reporting.py`.

Metric (current run)	Value
Coin exact-match accuracy	76.19% (80/105)
Value evaluated images	103/105
Value MAE	1.791 EUR per image
Value MSE	15.3455 EUR^2 per image

End-to-End Method Flow

How We Read Data (Code-backed)

- `ImageDataset.list_images()` recursively scans `data/images/` and keeps only valid extensions (`src/data/dataset.py`).
- Files are sorted by relative path so runs are deterministic and reproducible.
- Group aliases are normalized (`grp3` $\rightarrow$ `gp3`) before matching (`normalize_group_name`).
- `GroundTruthRepository` parses `RAW_ANNOTATIONS` and supports comma/dot decimals (`src/data/ground_truth.py`).
- CLI matches each image with GT by $(group, filename)$; missing GT is marked `SKIP_NO_GT` (`src/app/cli.py`).

Preprocessing Choices and Why

Step (implemented)	Reason in our pipeline
Letterbox to 640×480	Keep geometric scale consistent so Hough thresholds and radius sweeps are comparable across mixed image resolutions.
RGB $\rightarrow$ grayscale	Reduce color variability; keep intensity/edge information needed by Hough circles.
Gaussian or median blur	Suppress high-frequency texture noise before circle voting; reduce false circles from background patterns.
Auto `param1` from Scharr gradient percentile	Scene-adaptive Canny thresholding inside Hough; avoids brittle fixed-edge threshold on easy vs hard backgrounds.
Runtime normalization path	Current implementation keeps preprocessing compact: grayscale + blur for stable cross-group behavior.

Processing and Model Choices: Why These Methods

Method	Why we used it
`cv2.HoughCircles` + adaptive `minRadius` sweep	Coins are near-circular; sweep + geometry penalties (nested/intrusion checks) makes detection robust when one fixed radius fails.
Inner/Border HSV split + radial structure checks	Bimetal vs single-color requires radial evidence, not only global color. We add KMeans radial separation and radial step checks for this.
Family inference (`bronze/gold/bimetal`)	Constrains denomination search space early and reduces confusion between visually close coins.
Scale + family-wise radius clustering + priors	Maps pixel radii to denomination probabilities even when camera distance changes; combines size likelihood + cluster priors + center proximity.
Deterministic rules + debug overlays	Gives explainable decisions per coin and supports slide-level diagnostics for failures and successes.

Stage 1: Preprocessing

- Letterbox normalization to fixed canvas 640×480 (`letterbox_resize_to_canvas`).
- RGB $\rightarrow$ grayscale conversion.
- Blur mode configurable (`gauss` or `median`); active default is Gaussian.
- Effective detection path uses grayscale $+$ blur.
- Objective: stabilize Hough detection across heterogeneous image resolutions/backgrounds.

Stage 2: Circle Detection

- Core detector: `cv2.HoughCircles` in `CoinDetector`.
- `param1` is auto-estimated from Scharr gradient distribution.
- `minRadius` is not fixed:
  - ▷ sweep candidate range;
  - ▷ score each candidate with geometric penalties:
  - ▷ concentric duplicates, nested circles, intrusions, instability.
- Selection policy: plateau/stability voting, with fallback when no clean candidate exists.
- Output: detected circles + debug overlays + full sweep diagnostics.

Stage 3: Color Analysis (Inner/Border)

- For each detected circle, split into:
  - ▷ inner disk;
  - ▷ border ring.
- Compute robust HSV descriptors:
  - ▷ circular hue average (weighted),
  - ▷ median saturation/value,
  - ▷ hue/saturation/value deltas between inner and border.
- Add structure evidence:
  - ▷ radial 2-cluster KMeans consistency,
  - ▷ radial step-change score,
  - ▷ edge roughness score.
- Deterministic decision lattice: `one-color-like`, `bi-metal-like`, `uncertain`.

Stage 4: Denomination and Value Estimation

- Family inference per coin:
  - bronze / gold / bimetal / unknown.
- Scale estimation (px/mm):
  - uses bimetal references when available,
  - fallback consensus voting from all radius hypotheses.
- Family-wise radius clustering (1D KMeans + model selection).
- Probabilistic score fusion for candidate denominations:
  - size likelihood,
  - cluster mapping prior,
  - center proximity prior.
- Aggregate final denomination counts and total value in cents.

Case 1: gp1/25.png with Accuracy Values

Case 1 — Medium/difficult background, separated coins

Original image | Program output (final stage)

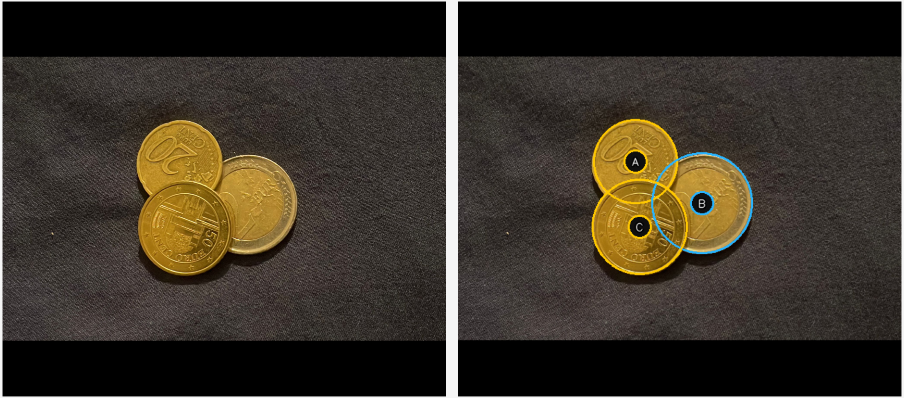
file=gp1/25.png | difficulty/status=hard/ERR | coin pred/true/diff=6/10/−4 | coin acc=60.0% | value pred/true/diff(EUR)=1.52/4.40/−2.88 | value acc=34.5%

Case 2: Overlap Coins, Successful Detection/Value

Case 2 — Overlap coins, successful detection/value

Original image

Program output (final stage)



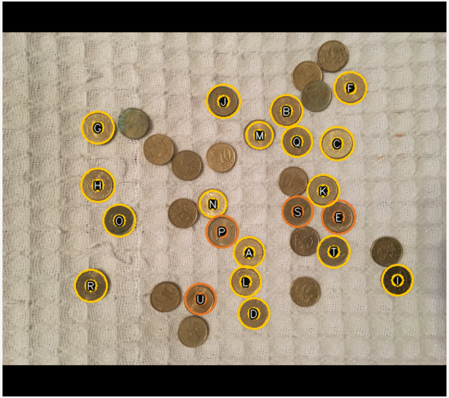
file=gp2/3.jpeg | difficulty/status=easy/OK | coin pred/true/diff=3/3/0 | coin acc=100.0% | value pred/true/diff(EUR)=2.70/2.70/+0.00 | value acc=100.0%

Case 3: Angle + Reflections (gp3/3_10.jpg)

Case 3 — Angle + reflections + adaptive radius sweep

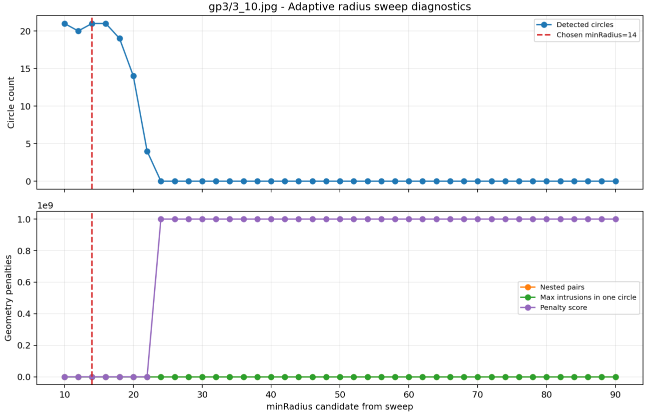
Original image

Program output (final stage)



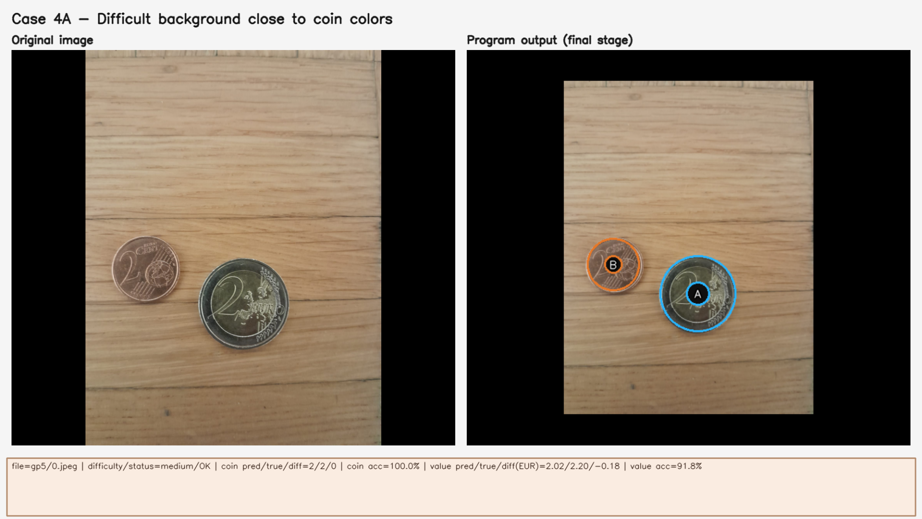
file=gp3/3_10.jpg | difficulty/status=hard/ERR | coin pred/true/diff=21/35/−14 | coin acc=60.0% | value pred/true/diff(EUR)=4.30/3.50/+0.80 | value acc=77.1%

Case 3 Diagnostics: Adaptive Radius Sweep



- We sweep candidate `minRadius` values.
- Candidates with nested circles or unstable geometry are penalized.
- We keep robust plateaus and select the best-supported regime.
- This directly addresses angle/reflection sensitivity for `gp3/3_10.jpg`.

Case 4A: Difficult Background Close to Coin Color (gp5/0.jpeg)

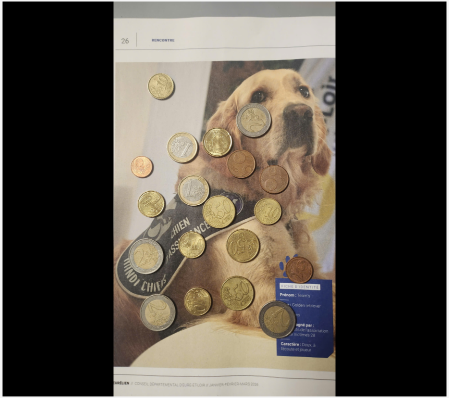



Case 4B — Difficult wood background, color confusion
Original image
Program output (final stage)
A
B
file=gp5/11.jpg | difficulty/status=hard/ERR | coin pred/true/diff=2/12/−10 | coin acc=16.7% | value pred/true/diff(EUR)=1.01/6.18/−5.17 | value acc=16.3%

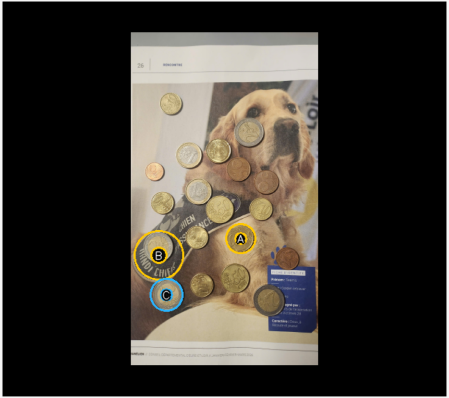
Case 4C: Hard Mixed Scene Near Coin Palette (gp5/13.jpg)

Case 4C — Printed image background, severe confusion

Original image



Program output (final stage)



file=gp5/13.jpg | difficulty/status=hard/ERR | coin pred/true/diff=3/19/−16 | coin acc=15.8% | value pred/true/diff(EUR)=2.30/12.33/−10.03 | value acc=18.7%

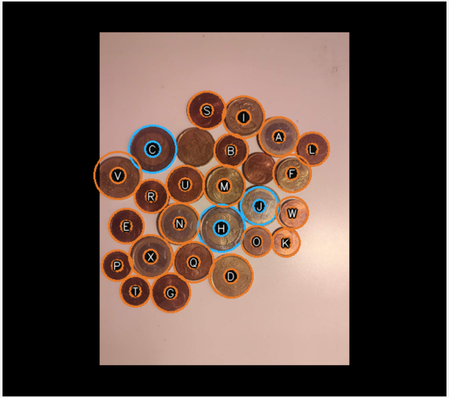
Case 4D: Very Hard Background (gp5/16.jpg)

Case 5: Coins Too Close to Each Other

Case 5 — Coins too close to each other

Original image

Program output (final stage)

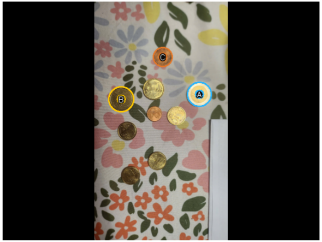


file=gp5/24.jpg | difficulty/status=hard/ERR | coin pred/true/diff=24/26/−2 | coin acc=92.3% | value pred/true/diff(EUR)=5.57/10.05/−4.48 | value acc=55.4%

Case 6 (assumed): circle-like background noise and color confusion

Original image

Program output (final stage)



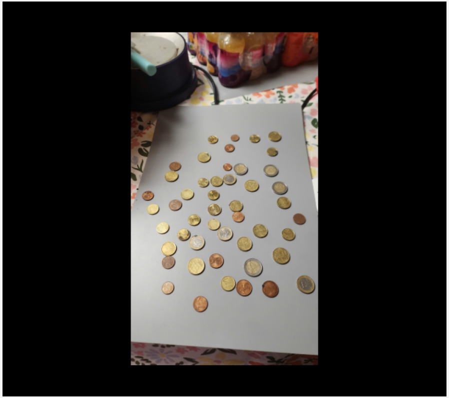
```
file: gp5/10.jpg
difficulty/status: hard / ERR
coin pred/true/diff: 3 / 9 / -6
coin ratio accuracy (pred vs true): 33.3%
value pred/true/diff (EUR): 2.55 / 3.13 / -0.58
value ratio accuracy (pred vs true): 81.5%
note: Assumption due to missing path in prompt
```

Failure Case: gp5/17.jpg (45° Camera Angle + Clutter)

Failure Case – gp5/17.jpg (45 degree angle + clutter)

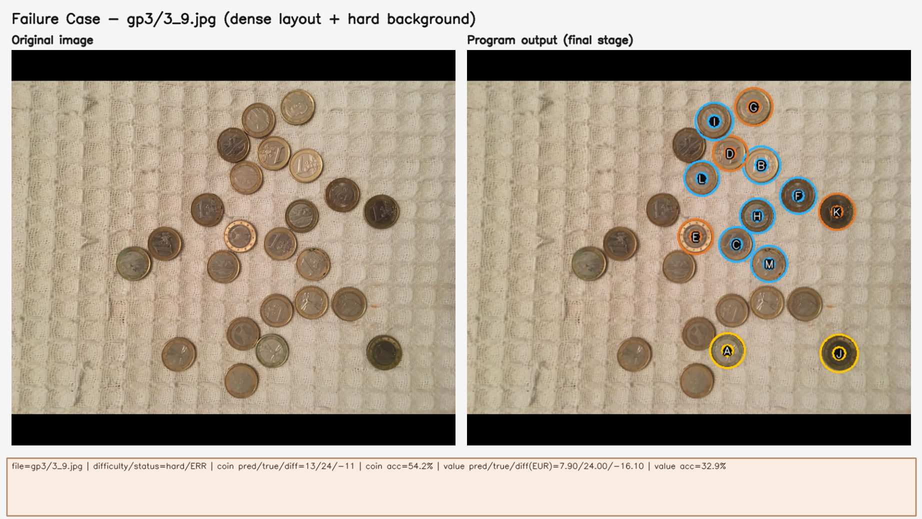
Original image

Program output (final stage)



file=gp5/17.jpg | difficulty/status=hard/ERR | coin pred/true/diff=0/48/−48 | coin acc=0.0% | value pred/true/diff(EUR)=0.00/18.20/−18.20 | value acc=0.0%

Failure Case: gp3/3_9.jpg (Dense Layout + Hard Background)

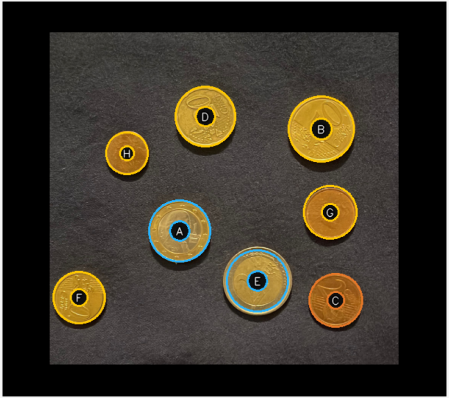


Normal/Easy Example: gp2/1.jpeg

Normal/easy example – gp2/1.jpeg

Original image

Program output (final stage)



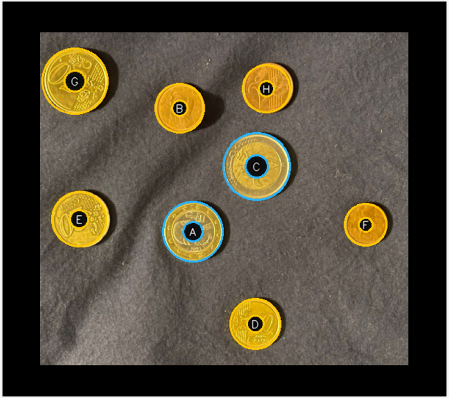
file=gp2/1.jpeg | difficulty/status=easy/OK | coin pred/true/diff=8/8/0 | coin acc=100.0% | value pred/true/diff(EUR)=4.55/–/– | value acc=–

Normal/Easy Example: gp2/4.jpeg

Normal/easy example – gp2/4.jpeg

Original image

Program output (final stage)



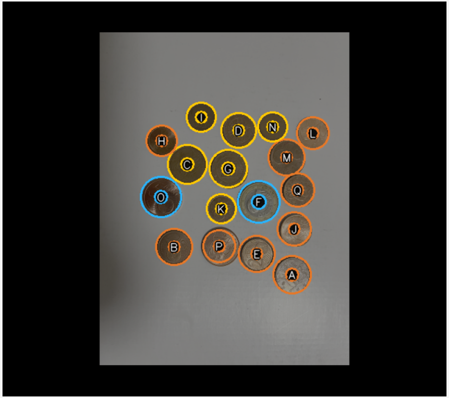
file=gp2/4.jpeg | difficulty/status=medium/OK | coin pred/true/diff=8/8/0 | coin acc=100.0% | value pred/true/diff(EUR)=4.10/3.86/+0.24 | value acc=93.8%

Normal/Easy Example: gp8/IMG_1143.png

Normal/easy example – gp8/IMG_1143.png

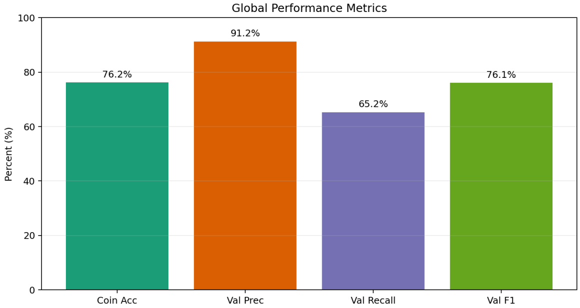
Original image

Program output (final stage)



file=gp8/IMG_1143.png | difficulty/status=hard/OK | coin pred/true/diff=17/17/0 | coin acc=100.0% | value pred/true/diff(EUR)=4.50/1.40/+3.10 | value acc=0.0%

Global Performance Snapshot



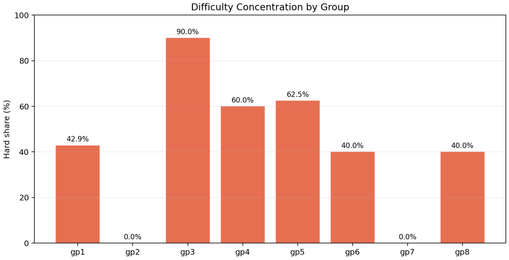
Observed strengths

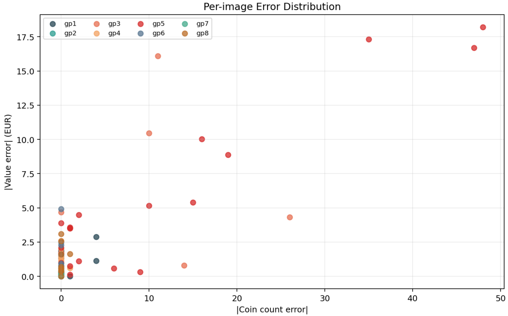
- Strong performance on clean groups (gp2, gp7, gp6).
- Stable full-run completion: 106/106 images processed.

Observed limits

- Coin exact-match accuracy is 76.19% overall.
- Hard groups (gp3, gp5) dominate large count/value errors.
- Value MAE rises sharply when many coins are missed in cluttered scenes.

Difficulty Distribution and Error Landscape





- Hard-share peaks in gp3 and gp5, matching visual clutter/overlap.
- Scatter confirms correlation: large coin-count errors tend to create large value errors.

Interpretation of Requested Cases

- **gp2/3.jpeg**: overlap is present, but boundary contrast is sufficient for correct count/value.
- **gp1/25.png**: coin spacing is good, but background texture/color biases value classification.
- **gp3/3_10.jpg**: angle and reflections reduce radius consistency and count accuracy.
- **gp5/17.jpg**: full failure $(0/48)$, with strong clutter and roughly $45°$ camera angle.
- **gp3/3_9.jpg**: failure from dense layout + textured background causing merge/miss errors.
- **gp5 hard set**: colors/shapes in background overlap coin cues, causing severe misses.
- **gp5/24.jpg**: close packing causes near-touching circles and merge/split ambiguity.

What We Learned

What worked

- Modular pipeline with clear debug artifacts per stage.
- Adaptive Hough setup improved robustness over fixed parameters.
- Inner/border analysis provided useful color-family cues.

Main failure modes

- Strong texture/color similarity between coins and background (notably in gp5).
- Perspective angle and reflections changing apparent radius.
- Dense layouts and close coins causing circle separation ambiguity.

Next Steps

- Add stronger segmentation before Hough (masking and region proposals).
- Improve overlap handling (multi-scale hypotheses + duplicate suppression).
- Add calibration with controlled reference objects.
- Explore learning-based detectors for hard groups while keeping explainability.
- Extend evaluation with per-denomination confusion analysis.

Reproducibility

Run full project evaluation

```
./.venv/bin/python main.py --no-view
```

Result CSV generated by the run

```
reports/evaluation_rows.csv
```

Build slides locally (XeLaTeX)

```
cd reports/overleaf_upload
xelatex -interaction=nonstopmode -halt-on-error slides.tex
xelatex -interaction=nonstopmode -halt-on-error slides.tex
```

Thank you